

Python Practical Questions for Lab File (CSE-104P)

Practical Question 1: Basic Building Blocks of a Python Program

Topic	Programming Tasks
Conditional Statements	Write a program that determines if a given year is a leap year.
Loops	Develop a program that prints out the Fibonacci sequence up to a specified number.

Note : Kindly do either of the two programs

Practical Question 2: Data Structures

Topic	Programming Tasks
Dictionaries	Create a program that counts the frequency of each word in a given text.
Tuples	Write a program that swaps the values of two variables using tuples.

Note : Kindly do either of the two programs

Practical Question 3: Object Oriented Programming

Topic	Programming Tasks
Classes	Class Definition: Define a class named BankAccount. Attributes: Include the following attributes in the constructor (__init__ method): - account_number: (String) A unique identifier for the account. - account_holder: (String) Name of the account holder. - balance: (Float) Initialize with a default value of 0.0 to represent the account balance. Methods: Add the following methods to the class: 1. deposit(amount): Accepts an amount and adds it to the balance. 2. withdraw(amount): Accepts an amount and subtracts it from the balance if sufficient funds are available. 3. check_balance(): Prints the current balance. 4. display_details(): Prints the account holder's details and current balance.
Objects	Create a simple Library Management System with three user roles: Student, Librarian, and Faculty. Each role should have specific use cases and required functions: 1. Student: - borrow_book(book): Method to borrow a book, updating its status to "borrowed". - return_book(book): Method to return a borrowed book, updating its status to "available". - check_due_date(book): Method to view the due date of a specific borrowed book. 2. Librarian: - add_book(book): Method to add new books to the library inventory. - remove_book(book): Method to remove books from the library inventory. - view_borrowed_books(): Method to list all currently borrowed books and their borrowers. 3. Faculty: - reserve_book(book): Method to reserve a book for future use, updating its status to "reserved". - extend_due_date(book, additional_days): Method to extend the due date of a borrowed book by a specified number of days. - suggest_new_book(book): Method to suggest a new book for the library to acquire.

Note : Kindly do either of the two programs

Practical Question 4: PyQt (Graphical User Interfaces)

Topic	Programming Tasks
GUI Design & Event Handling	Question 1: Interactive Temperature Conversion Application Develop a desktop application using the <code>PyQt5</code> framework that serves as a bidirectional temperature converter. The application window must contain a <code>QVBoxLayout</code> or <code>QFormLayout</code> to organize the following widgets: 1. A <code>QLineEdit</code> for numeric input. 2. A <code>QComboBox</code> or two <code>QRadioButton</code> widgets to select the conversion direction (Celsius to Fahrenheit or Fahrenheit to Celsius). 3. A <code>QPushButton</code> labeled "Convert" that, when clicked, triggers a Python function to perform the calculation using the standard formulas: $F = (C * 9/5) + 32$ and $C = (F - 32) * 5/9$. 4. A <code>QLabel</code> to display the final result formatted to two decimal places. Requirements: Use <code>sys.argv</code> to initialize the <code>QApplication</code> and ensure the application remains responsive during user interaction.
	Question 2: Design a functional calculator using <code>QGridLayout</code>. The application must feature a display area (non-editable <code>QLineEdit</code>) and a set of buttons for digits (0-9) and basic operators (+, -, *, /). Implement the "Clear" and "Equals" functionality, ensuring that mathematical operations are evaluated correctly.
	Question 3: Create a comprehensive GUI wrapper for either the BankAccount or the Library Management System developed in Practical Question 3. The application must provide a visual interface to perform core logic. Requirement: Integrate data persistence using the <code>csv</code> or <code>json</code> module so that the data is retained when the application is restarted.

Note: Kindly do either of the three programs.

Practical Question 5: NumPy (Numerical Computing)

Topic	Programming Tasks
Array Math & Logic	Question 1: Multi-Dimensional Array Analysis and Attribute Extraction Write a Python script using the NumPy library to perform a series of matrix operations. 1. Create a 3 x 3 NumPy array (matrix) and populate it with integer values ranging from 1 to 20. 2. Print the Shape, Size, and Data Type of the array. 3. Generate and display the Transpose of the matrix. 4. Calculate the Global Sum of all elements and the Mean value across axis=1 (row-wise). 5. Use boolean indexing to identify and print all elements in the matrix that are greater than 10. Requirements: Use numpy.array() for initialization and numpy.mean() for statistical calculations.
	Question 2: Develop a script that performs algebraic operations on two N x N matrices. The program must: 1. Perform a valid Matrix Multiplication of the two matrices. 2. Calculate the Determinant of the resulting matrix. 3. If the matrix is non-singular, compute its Inverse. Ensure the code handles potential errors, such as dimension mismatches.
	Question 3: Image Transformation Engine Develop a program that treats a grayscale image as a 2D NumPy array to perform basic image processing. The program must: 1. Load an image file into a NumPy array. 2. Implement a function to Flip the image horizontally or vertically using array slicing. 3. Implement a Brightness Filter that increases/decreases pixel values by a user-defined scalar. 4. Apply a Binary Threshold: Every pixel above a value of 127 becomes 255 (white), and below becomes 0 (black).

Note: Kindly do either of the three programs.

Practical Question 6: SciPy (Scientific Tools)

Topic	Programming Tasks
Physics & Root Finding	Question 1: Calculate Gravitational and Electrostatic force Implement a modular Python program using the <code>scipy.constants</code> library to calculate forces between objects based on physical laws. 1. Define a function <code>calculate_gravity(m1, m2, r)</code> that uses the Newtonian constant of gravitation (G) to return the force $F = G * (m1 * m2) / r^2$. 2. Define a function <code>calculate_electrostatic(q1, q2, r)</code> that uses the vacuum permittivity constant (ϵ_0) to calculate the Coulomb constant ($k_e = 1 / (4 * \pi * \epsilon_0)$) and return the force $F = k_e * (q1 * q2) / r^2$. 3. The program should accept user inputs for mass/charge and distance, then output the result in Scientific Notation with two decimal places. Requirements: Import <code>pi</code> , <code>G</code> , and <code>epsilon_0</code> directly from the <code>scipy.constants</code> module.
	Question 2: Implement a numerical solver using <code>scipy.optimize</code> . The program should find the roots of a non-linear transcendental equation (e.g., $x + 2 * \cos(x) = 0$). The output should include the root value and a confirmation that the function evaluates to approximately zero at that point.
	Question 3: Digital Signal/Audio Analysis Write a script that utilizes <code>scipy.io.wavfile</code> and <code>scipy.fft</code> to analyze a recorded audio sample. The program should read a <code>.wav</code> file, convert the signal into the frequency domain using a Fast Fourier Transform (FFT), and identify the dominant frequency (pitch) in Hertz.

Note: Kindly do either of the three programs.

Practical Question 7: Matplotlib (Visualization)

Topic	Programming Tasks
Visual Analytics	Question 1: Professional Revenue Trend Visualization Develop a Python script using <code>Matplotlib</code> to visualize a dataset representing the monthly revenue of a business. Input Data: Define two lists: <code>months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']</code> and <code>revenue = [5.2, 7.8, 6.5, 9.1, 12.4, 11.5]</code>. Plot Requirements: - Create a Line Plot with a red dashed line (<code>linestyle='--'</code>) and circular markers (<code>marker='o'</code>). - Add a descriptive Title ("Start-up Revenue Growth - H1 2026"). - Label the X-axis as "Month" and the Y-axis as "Revenue in Lakhs (INR)". - Enable the background Grid for readability and add a Legend identifying the data series. - Use <code>plt.show()</code> to render the plot.
	Question 2: Generate a multi-line graph comparing the historical prices of USD (vs INR) and Gold (per 10g) from 1970 to 2020. The plot must use different colors for each asset and include a clear legend.
	Question 3: Live System Telemetry Dashboard Develop an interactive dashboard using the <code>animation</code> module of Matplotlib and the <code>psutil</code> library. The application must visualize real-time CPU Usage and Available Memory on a live-updating line chart. Ensure the X-axis represents a sliding window of time.

Note: Kindly do either of the three programs.

Practical Question 8 : Interactive Python

Write a IPython script to demonstrate five IPython commands in a IPython Notebook (.ipynb file)